

module5_lab3

October 4, 2026

```
[1]: import pandas as pd
df=pd.read_csv("Downloads/custdata.tsv", sep="\t")
```

```
[2]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 11 columns):
#   Column          Non-Null Count  Dtype
---  -
0   custid          1000 non-null   int64
1   sex             1000 non-null   str
2   is.employed     672 non-null    object
3   income          1000 non-null   int64
4   marital.stat    1000 non-null   str
5   health.ins      1000 non-null   bool
6   housing.type    944 non-null    str
7   recent.move     944 non-null    object
8   num.vehicles    944 non-null    float64
9   age            1000 non-null   float64
10  state.of.res    1000 non-null   str
dtypes: bool(1), float64(2), int64(2), object(2), str(4)
memory usage: 115.8+ KB
```

```
[3]: df.info
```

```
[3]: <bound method DataFrame.info of          custid sex is.employed  income
marital.stat  health.ins  \
0          2068   F         NaN   11300          Married      True
1          2073   F         NaN     0          Married      True
2          2848   M         True   4500      Never Married     False
3          5641   M         True  20000      Never Married     False
4          6369   F         True  12000      Never Married      True
..          ...   ..         ...   ...          ...          ...
995      1411132   F        False  46000  Divorced/Separated      True
996      1411860   F         NaN   5400          Widowed      True
997      1412161   M         True  38500          Married      True
998      1412971   M         NaN  43400          Married      True
```

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999  1414286  F      False  20900      Married      False

      housing.type recent.move num.vehicles  age  state.of.res
0    Homeowner free and clear  False      2.0  49.0    Michigan
1              Rented      True      3.0  40.0     Florida
2              Rented      True      3.0  22.0     Georgia
3    Occupied with no rent  False      0.0  22.0    New Mexico
4              Rented      True      1.0  31.0     Florida
..          ...          ...          ...  ...  ...
995              Rented      True      1.0  56.0     Florida
996              NaN      NaN      NaN  89.0    Pennsylvania
997              Rented      True      1.0  29.0     Georgia
998  Homeowner free and clear  False      1.0  88.0      Ohio
999              Rented      False     2.0  36.0    New York

```

```
[1000 rows x 11 columns]>
```

```
[4]: df.isnull().sum()
```

```

[4]: custid      0
     sex        0
     is.employed  328
     income      0
     marital.stat  0
     health.ins   0
     housing.type  56
     recent.move  56
     num.vehicles  56
     age         0
     state.of.res  0
     dtype: int64

```

```
[5]: df.describe()
```

```

[5]:
count    custid      income  num.vehicles      age
mean    6.984997e+05  53504.771000    1.916314    51.699815
std      4.135083e+05  65478.065729    1.101618    18.863433
min      2.068000e+03  -8700.000000    0.000000    0.000000
25%      3.456668e+05  14600.000000    1.000000    38.000000
50%      6.934030e+05  35000.000000    2.000000    50.000000
75%      1.044606e+06  67000.000000    2.000000    64.000000
max      1.414286e+06  615000.000000    6.000000   146.680197

```

```

[6]: #markdown cell
     #there are no missing numbers for age,income, but 56 are missing from vehicles

```

#there is a discrepancy to the income as there is a low min of -8700 so it
↳ could be a typo or mistake
#there is a discrepancy in the age as well with customers being babies and
↳ older data point at 146 years old

```
[7]: cols = ["age", "income", "num.vehicles"]
```

```
[8]: corr_before=df[cols].corr()
```

```
[9]: print(corr_before)
```

	age	income	num.vehicles
age	1.000000	0.027429	-0.063666
income	0.027429	1.000000	0.139198
num.vehicles	-0.063666	0.139198	1.000000

```
[10]: cleaned_df = df[(df["age"] >= 18) & (df["age"] <= 100)]
cleaned_df = cleaned_df[cleaned_df["income"] >= 0]
```

```
[11]: #edits
#removed age where there was possible errors, removing anyone under the age of
↳ 18 an above 100 years old
#removed income of anyone less than 0, or making negative income, for analysis
```

```
[12]: cleaned_df.describe()
```

```
[12]:
```

	custid	income	num.vehicles	age
count	9.880000e+02	988.000000	932.000000	988.000000
mean	6.958594e+05	53129.879555	1.914163	51.215587
std	4.132938e+05	65200.621337	1.103229	17.363014
min	2.068000e+03	0.000000	0.000000	18.000000
25%	3.430412e+05	14575.000000	1.000000	38.000000
50%	6.886580e+05	35000.000000	2.000000	50.000000
75%	1.042425e+06	66250.000000	2.000000	63.000000
max	1.414286e+06	615000.000000	6.000000	93.000000

```
[13]: #verified cleaned_df has removed age values above 100 and below 18, thus
↳ removing 12 rows from ages
#verified income also removed incomes that were impossible and was removed with
↳ age as the total numbe for income match age count
```

```
[14]: cols = ["age", "income", "num.vehicles"]
corr_after=cleaned_df[cols].corr()
print(corr_after)
```

	age	income	num.vehicles
age	1.000000	0.008881	-0.066295
income	0.008881	1.000000	0.137859

```
num.vehicles -0.066295  0.137859      1.000000
```

```
[15]: #Correlation statement
# there is an r value of .137 between income and number of vehicles pointing to
    ↳ higher incomes to having more cars.
# the r value of .008 between income and age shows theres no relationship
    ↳ between ones age and income made.
```